

graphing

October 24, 2023

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[20]: import matplotlib.pyplot as plt
import numpy as np
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[51]: def gen_graphs():

    transitions = []
    c_elapsed_time = []
    java_elapsed_time = []
    idx = 0

    with open('../result.txt', 'r') as file:
        file_contents = file.read()

    lines = file_contents.split('\n')

    for i in range(0, len(lines), 3):
        transitions.append(lines[i])

    for i in range(1, len(lines), 3):
        c_elapsed_time.append(float(lines[i]))

    for i in range(2, len(lines), 3):
        java_elapsed_time.append(float(lines[i]))

    transitions.pop()
    transitions = [float(x) for x in transitions]
    java_elapsed_time = [float(x) for x in java_elapsed_time]
    c_elapsed_time = [float(x) for x in c_elapsed_time]
    java_elapsed_time_array = np.array(java_elapsed_time)
    c_elapsed_time_array = np.array(c_elapsed_time)
    data_overhead = java_elapsed_time_array - c_elapsed_time_array
    figure, axis = plt.subplots(1,2,figsize=(14, 6))
    # axis[0,0].rc('font', size=14)
    axis[0].set_title("Transitions between Java code and Native Code (10 MB)")
    axis[0].set_xlabel("No. of Transitions")
    axis[0].set_ylabel("Time Elapsed (in seconds)")
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axis[0].plot(transitions,java_elapsed_time)

# plt.figure(2)
axis[1].plot(transitions,data_overhead)
axis[1].set_title("Cost of Data Transfer between Java Code and Native Code_10 MB")
axis[1].set_xlabel("No. of Transitions")
axis[1].set_ylabel("Time Elapsed (in seconds)")
plt.show()

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```

[52]: % bash
cd ..
python3 gen_file.py 34800 300 > hashing.txt
make -s

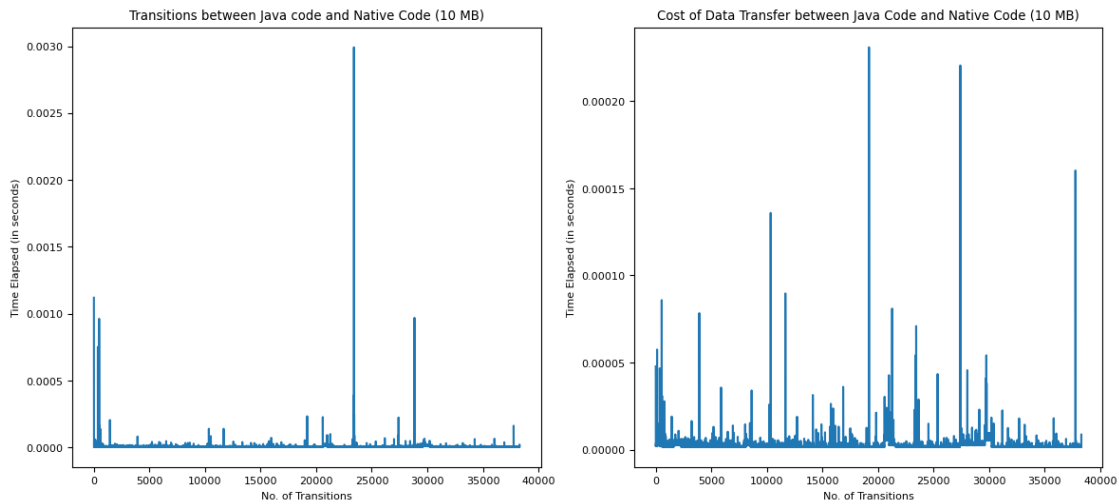
```

1 10 MB

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[53]: gen_graphs()

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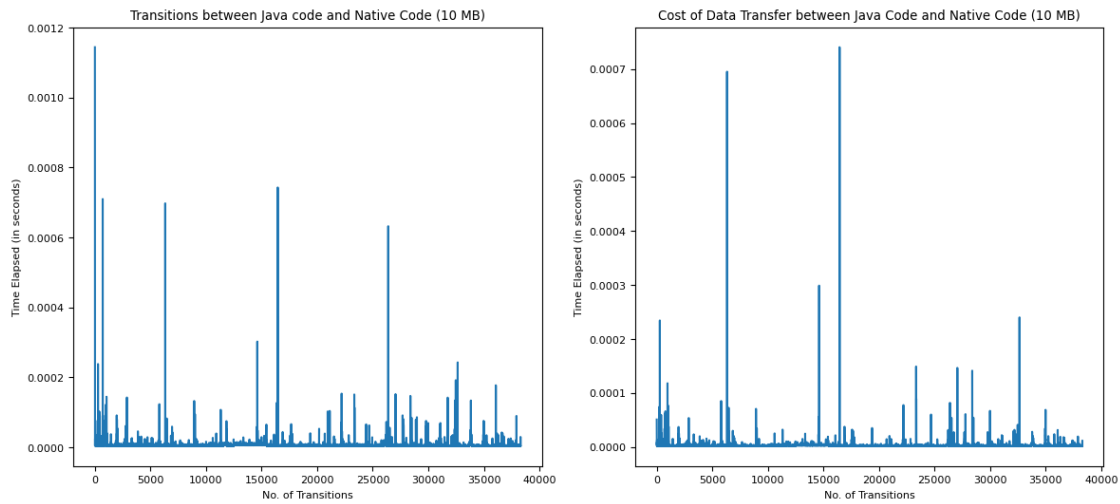
```

[54]: % bash
cd ..
python3 gen_file.py 348 300 > hashing.txt
make -s

```

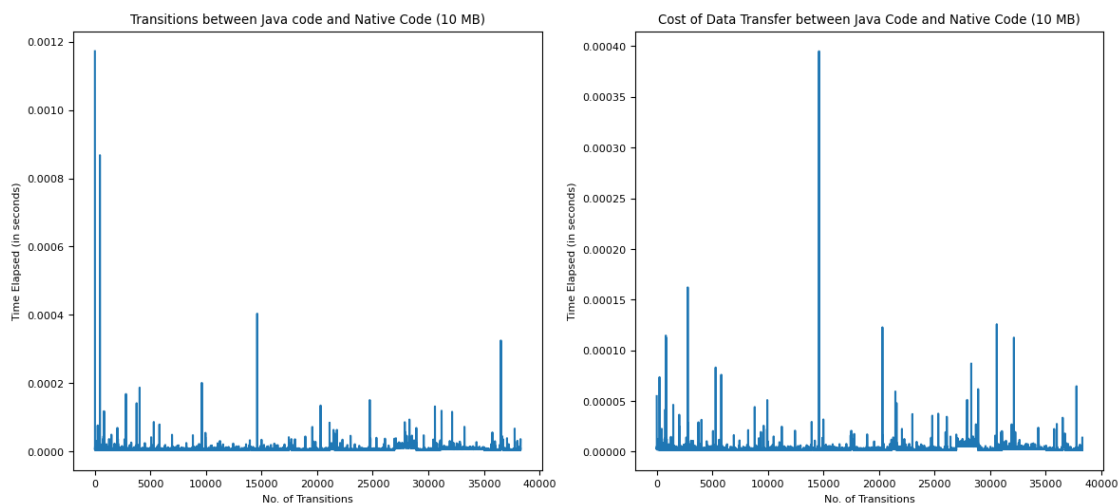
2 1 MB

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[55]: gen_graphs()
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```
[56]: % bash
      cd ..
      python3 gen_file.py 1005 100 > hashing.txt
      make -s
```

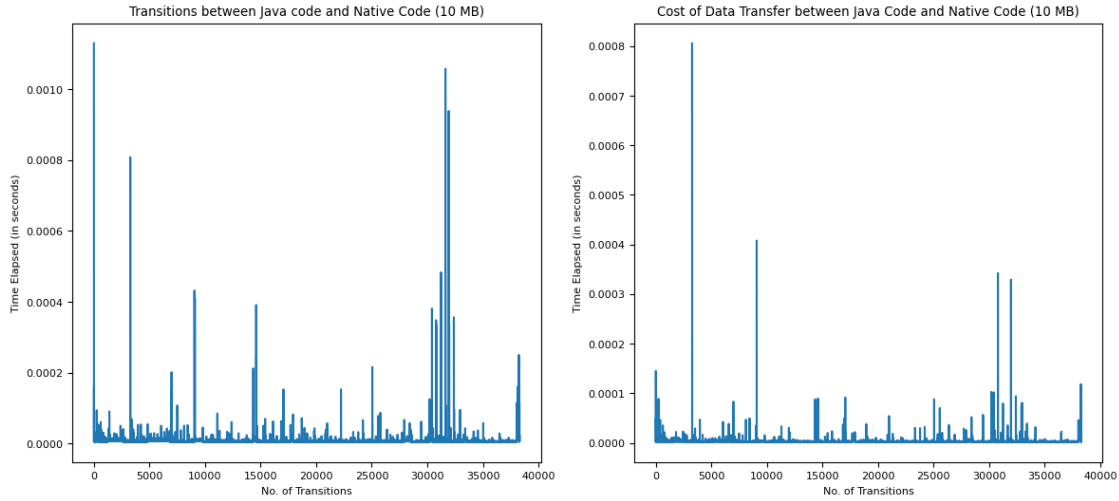
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[57]: gen_graphs()
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```
[58]: % bash
      cd ..
```

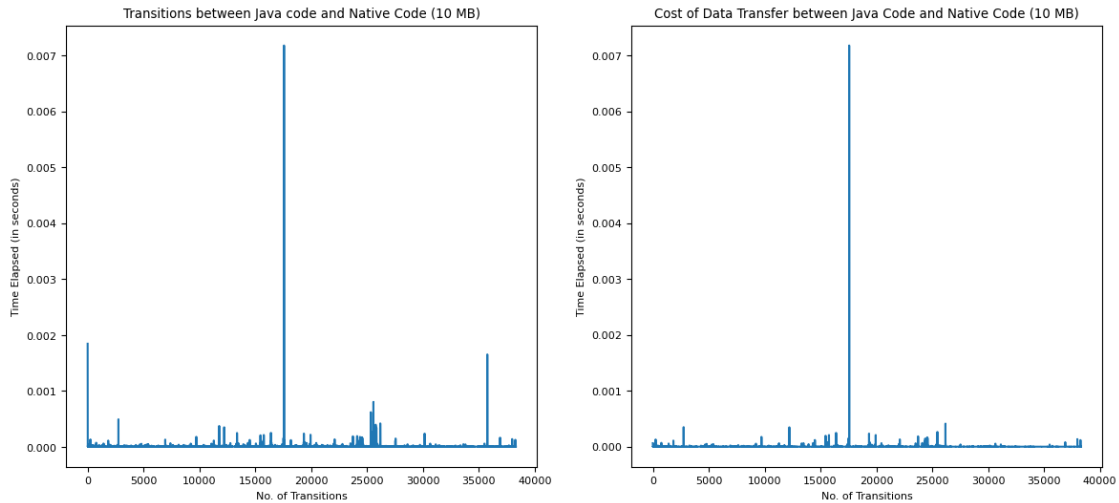
```
python3 gen_file.py 101 100 > hashing.txt
make -s
```

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[59]: gen_graphs()
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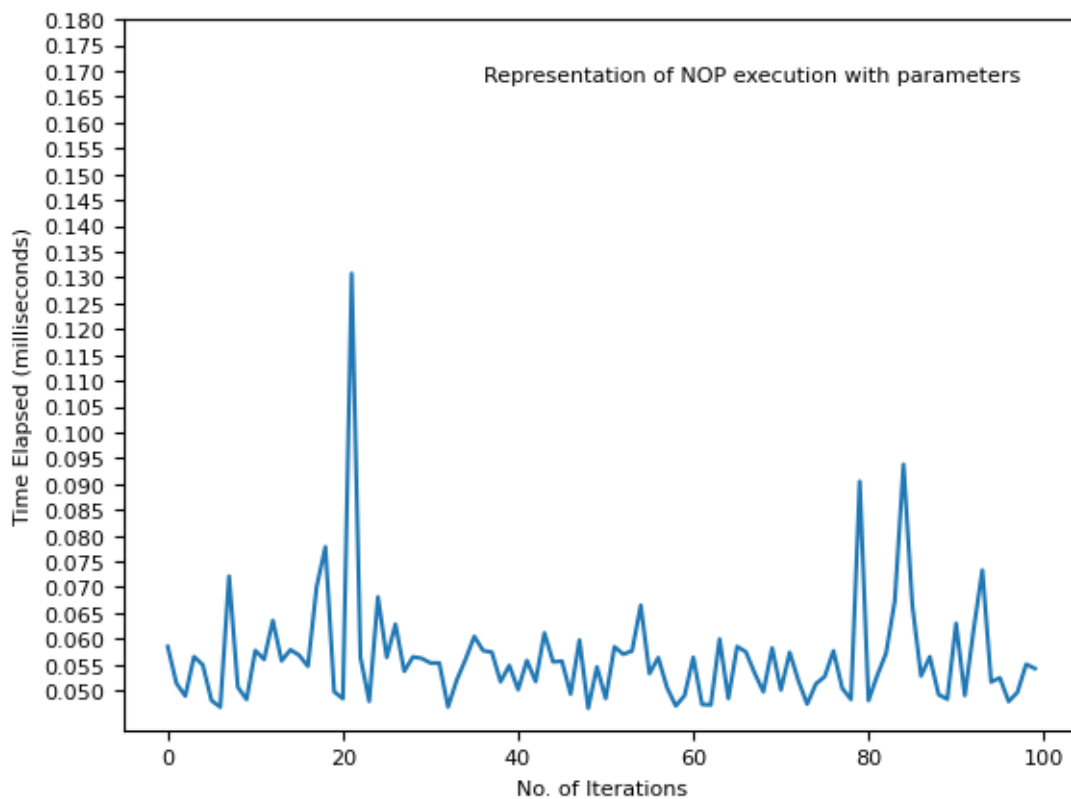


```
[60]: % bash
cd ..
python3 gen_file.py 102 90 > hashing.txt
make -s
```

```
[61]: gen_graphs()
```



```
[62]: x_axis = []
y_axis = []
idx = 0
with open('../NOP/NOP_params/visualization/exec.txt', 'r') as file:
    for line in file:
        y_axis.append(float(line.strip()))
        x_axis.append(idx)
        idx+=1
plt.rc('font', size=8)
plt.plot(x_axis,y_axis)
plt.yticks(np.arange(0.05, max(y_axis)+0.05, step=0.005))
plt.xlabel('No. of Iterations')
plt.ylabel('Time Elapsed (milliseconds)')
plt.text(36, 0.168, 'Representation of NOP execution with parameters')
plt.show()
```

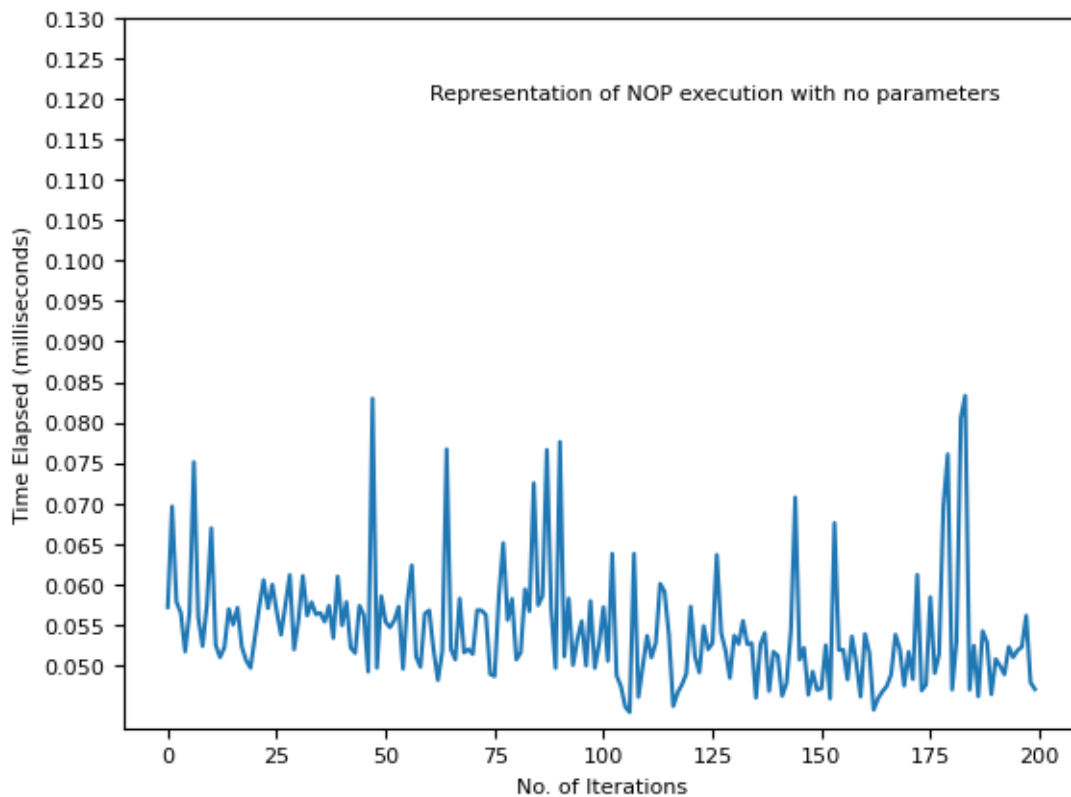


```
[63]: x_axis = []
y_axis = []
idx = 0
with open('../NOP/NOP_noparams/visualization/exec.txt', 'r') as file:
    for line in file:
```

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        y_axis.append(float(line.strip()))
        x_axis.append(idx)
        idx+=1
plt.rc('font', size=8)
plt.plot(x_axis,y_axis)
plt.yticks(np.arange(0.05, max(y_axis)+0.05, step=0.005))
plt.xlabel('No. of Iterations')
plt.ylabel('Time Elapsed (milliseconds)')
plt.text(60, 0.12, 'Representation of NOP execution with no parameters')
plt.show()

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[]: